

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

PHASE 3: PROJECT DESIGN

1. System Architectural Design & Tier Topology

The technical design of the Auto Ticket Classification system is built on ServiceNow's enterprise platform architecture. The solution divides cleanly into a Presentation Layer (UI/Form Designer), Application Logic Layer (Flow Designer Workflow Studio), and Relational Database Storage Layer.

- **Presentation Layer:** Dynamic Form Designer interface with field-level choice constraints and contextual menus.
- **Business Logic Engine:** Event-driven Flow Designer process executing asynchronous database updates and transaction logging.
- **Persistence Layer:** Custom relational database table `u\_incident\_workflow` inheriting core ServiceNow system dictionary attributes.

2. Database Schema & Data Dictionary Specification

The data model centers on the custom table `Incident WorkFlow` (`u\_incident\_workflow`). The table schema is structured to capture mandatory audit attributes, user reference keys, and categorical choice metadata.

Field Label	Database Column Name	Data Type	Length / Reference Table	Behavior / Choice Values
Number	u_number	Auto Number	40 Characters	Read-only, Prefix: INC, Seed: 500, Digits: 5
Caller	u_caller	Reference	sys_user (32 chars)	Mandatory, links to student or faculty account
Category	u_category	Choice	40 Characters	Choices: Network, Hardware, Access, Performance

Subcategory	u_subcategory	Choice	40 Characters	Dependent choices based on Category selection
Short Description	u_short_description	String	160 Characters	Mandatory text string evaluated by Flow Designer
Description	u_description	String	4000 Characters	Optional long-form narrative describing incident details
State	u_state	Choice	40 Characters	Choices: New, In progress, On hold, Resolved, Closed
Assigned Group	u_assigned_group	Reference	sys_user_group (32 chars)	Optional routing assignment bucket
Assigned To	u_assigned_to	Reference	sys_user (32 chars)	Optional direct technician assignment

3. Dependent Choice Hierarchy & Normalization

To eliminate invalid data entry and maintain referential clarity across reporting dashboards, the Subcategory field is configured with a strict dictionary-level dependency on the Category field. If a user selects 'Network', only network-related subcategories appear in the UI.

Parent Category Label	Backend Value	Child Subcategory Label	Child Backend Value	Dependent Value Filter
Network	network	Wi-Fi	wi-fi	network
Hardware	hardware	Projector	projector	hardware
Access	access	Forgot Password	forgot password	access
Performance	performance	Slow Computer	slow computer	performance